Learning Spatial-Temporal Coherent Correlations for Speech-Preserving Facial Expression Manipulation

Tianshui Chen, Jianman Lin, Zhijing Yang, Chunmei Qing, *Member, IEEE*, Guangrun Wang, and Liang Lin, *Fellow, IEEE*

***Abstract*—Speech-preserving facial expression manipulation (SPFEM) aims to modify facial emotions while meticulously maintaining the mouth animation associated with spoken content. Current works depend on inaccessible paired training samples for the person, where two aligned frames exhibit the same speech content yet differ in emotional expression, limiting the SPFEM applications in real-world scenarios. In this work, we discover that speakers who convey the same content with different emotions exhibit highly correlated local facial animations in both spatial and temporal spaces, providing valuable supervision for SPFEM. To capitalize on this insight, we propose a novel spatial-temporal coherent correlation learning (STCCL) algorithm, which models the aforementioned correlations as explicit metrics and integrates the metrics to supervise manipulating facial expression and meanwhile better preserving the facial animation of spoken content. To this end, it first learns a spatial coherent correlation metric, ensuring that the visual correlations of adjacent local regions within an image linked to a specific emotion closely resemble those of corresponding regions in an image linked to a different emotion. Simultaneously, it develops a temporal coherent correlation metric, ensuring that the visual correlations of specific regions across adjacent image frames associated with one emotion are similar to those in the corresponding regions of frames associated with another emotion. Recognizing that visual correlations are not uniform across all regions, we have also crafted a correlation-aware adaptive strategy that prioritizes regions that present greater challenges. During SPFEM model training, we construct the spatial-temporal coherent correlation metric between corresponding local regions of the input and output image frames as additional loss to supervise the generation process. We conduct extensive experiments on various datasets, and the results demonstrate the effectiveness of the proposed STCCL algorithm.**

***Index Terms*—Facial expression manipulation, talking head manipulation, correlation learning, spatial-temporal correlation learning.**

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Tianshui Chen and Zhijing Yang are with the Guangdong University of Technology, Guangzhou 510006, China (e-mail: tianshuichen@gmail.com; yzhj@gdut.edu.cn).

Jianman Lin and Chunmei Qing are with the South China University of Technology, Guangzhou 510006, China (e-mail: linjianmancjx@gmail.com; qchm@scut.edu.cn).

Guangrun Wang and Liang Lin are with Sun Yat-sen University, Guangzhou 510275, China (e-mail: wanggrun@gmail.com; linliang@ieee.org).

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I. Introduction

Speech-preserving Facial Expression Manipulation (SPFEM) aims to modify facial emotional states in static images or dynamic videos while rigorously maintaining the mouth animation associated with the spoken content. This technology holds the potential to enhance digital human expressiveness, thereby benefiting a wide range of applications, from virtual avatars to professional film and television production. For instance, capturing the precise emotional nuance of an actor often necessitates exhaustive efforts and repeated retakes during shooting. In contrast, a robust SPFEM system offers a transformative solution for the post-production stage, enabling the flexible modification of facial performance without the need for costly reshoots. Consequently, the development of such high-fidelity systems is urgently demanded.

Current SPFEM literature either predominantly employs previous face reenactment algorithms [1], [2] or harnesses decoupled semantic representations equipped with cyclic consistency [3], [4]. The former category of works [1], [2] typically manipulates facial expressions through the exchange of latent codes [5] or facial action units [6], and employs the reference images as surrogate labels to construct frame-by-frame reconstruction supervision. However, these surrogate images are not perfect representations of the desired outcomes and the reliance on them may lead to generating sub-optimal results. The latter approach [4] posits a one-to-one correspondence between images exhibiting varied emotional expressions and employs cyclic consistency for paired supervision. Despite achieving better performance, the global cyclical consistency constraint makes it difficult to capture alterations in fine-grained facial information under different emotions. Consequently, these methods face a dilemma: they either fail to accurately translate the intended emotions (as illustrated in the first and third examples in Fig. 2) or fail to maintain the original mouth animation tied to the spoken content (as evidenced in the first and second examples in Fig. 2).

Notably, this bottleneck is not confined to specific SPFEM architectures but persists even in the broader field of emotional talking head generation. State-of-the-art paradigms, spanning from semantically disentangled representation learning (e.g., StyleTalk [7], EAT [8]) to emerging diffusion-based generative frameworks (e.g., DICE-Talk [9]), predominantly rely on self-driven training strategies to manipulate facial expressions while preserving speech alignment. Although these methods achieve impressive texture fidelity, whether through implicit feature disentanglement or probabilistic conditional generation, they

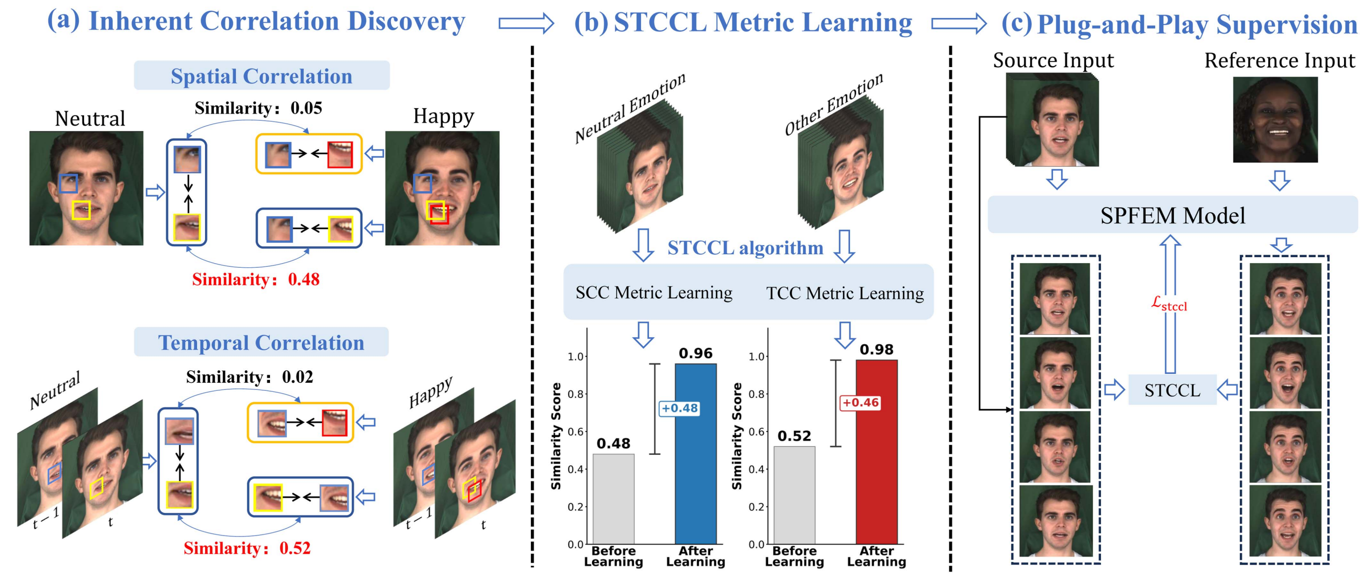

Fig. 1. Overview of the proposed Spatial-Temporal Coherent Correlation Learning (STCCL) framework. **(a) Inherent Correlation Discovery:** We uncover a core observation that a single speaker articulating identical speech content across varied emotional states exhibits highly correlated local facial animations in both spatial and temporal dimensions (see detailed statistical validation in Fig. 3). **(b) STCCL Metric Learning:** Leveraging limited paired training data, we explicitly model these intrinsic associations by training a Spatial Coherent Correlation (SCC) metric and a Temporal Coherent Correlation (TCC) metric, which effectively amplifies the discriminability of latent correlations (detailed network architectures and formulations are illustrated in Fig. 5). **(c) Plug-and-Play Supervision:** The learned STCCL serves as an architecture-agnostic auxiliary supervision ($\mathcal{L}_{stccl}$) for mainstream SPFEM models. By enforcing source-consistent spatial-temporal constraints between the source input and generated output, it ensures high-fidelity speech preservation amidst vivid emotional manipulation (see Fig. 4 for the complete integration workflow).

Fig. 2. Several examples are generated by the current advanced NED with and without the proposed STCCL algorithm. Incorporating the STCCL can better manipulate the expressions and meanwhile preserve mouth shapes.

inevitably encounter an inherent trade-off between emotional expressiveness and articulatory precision. Specifically, relying on implicit feature disentanglement often results in articulatory ambiguity during large-scale geometric deformations, while the stochastic nature of the diffusion denoising process inevitably compromises the precise mouth shape control essential for fine-grained lip synchronization. Consequently, ensuring fine-grained lip synchronization amidst vivid emotional expression remains an open challenge.

To overcome this ubiquitous limitation across varying paradigms and establish accurate supervisory guidance without paired data, we move beyond direct pixel supervision and investigate the intrinsic structural consistency of facial dynamics. **We uncover a core observation: a single speaker articulating identical content across varied emotional states exhibits strong correlations in local facial animations within each image and across consecutive images (as illustrated in Fig. 1(a)).** Specifically, in the context of speech-preserving emotion generation, we hypothesize that these correlations persist between spatially adjacent and temporally consecutive regions of the input frames and their corresponding counterparts in the output frames. To validate this, we quantitatively analyze the spatial and temporal correlation coefficients. As depicted in Fig. 3, we observe consistently high correlation coefficients for corresponding local regions, whereas non-corresponding regions demonstrate values approaching zero. This empirical evidence suggests that these intrinsic correlations can serve as a pseudo-paired signal, offering the explicit structural constraints needed to resolve the ambiguity inherent in unsupervised generation.

To this end, we propose a novel spatial-temporal coherent correlation learning algorithm (STCCL) that learns spatial correlations between adjacent local regions within each image and temporal correlations between the corresponding regions

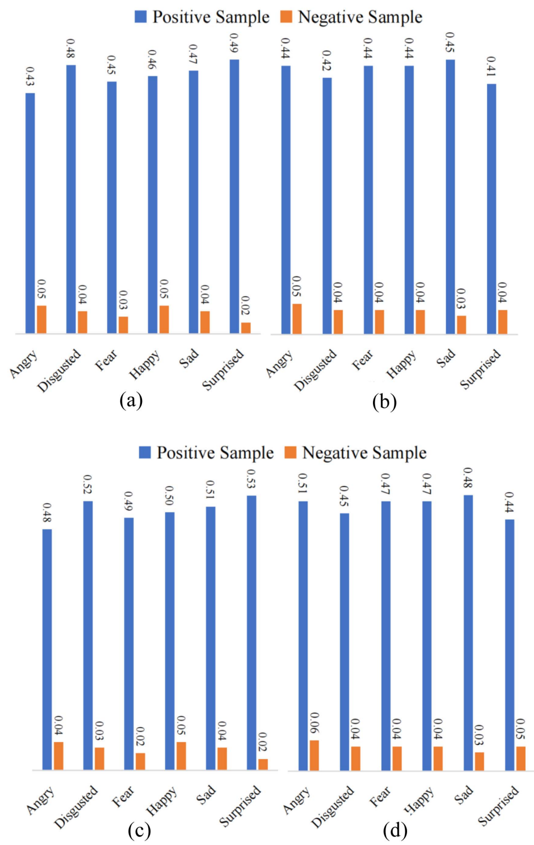

Fig. 3. Statistical validation of the key observation in Fig. 1(a). We compare the similarities of positive (corresponding) vs. negative (non-corresponding) region pairs across varying emotions. (a)-(b) evaluate spatial correlations using visual disparities and correlation matrices, respectively, while (c)-(d) evaluate temporal correlations. The significantly higher similarities in positive pairs confirm the inherent spatial-temporal coherence during identical speech, directly motivating STCCL to exploit this consistency as a supervision signal.

across consecutive frames. The algorithm then incorporates these learned correlations as additional guidance to supervise facial expression manipulation in a difficulty-aware manner. Formally, we leverage visual disparities and correlation matrices to characterize correlations between local regions, since local motion disparities and these correlation structures are critical for realistic facial animations. We learn a spatial coherent correlation (SCC) metric to ensure that correlation patterns among adjacent local regions in an image with one emotion are similar to those in the corresponding image with another emotion. Similarly, we introduce a temporal coherent correlation (TCC) metric to ensure that correlation patterns for specific regions across consecutive frames of one emotion are similar to those in the corresponding regions across frames of another emotion. To account for the varying complexity of different facial regions, we further propose a correlation-aware adaptive weighting strategy that assigns higher weights to more challenging regions and correspondingly lower weights to less complex regions. During SPFEM model training, we establish dense correlation mappings between local regions of the input and output images and use both spatial and temporal correlation metrics as an auxiliary supervisory signal. After training on paired data, STCCL can be applied in a plug-and-play manner to provide this form of supervision to new subjects (as shown in Fig. 1(c)), thereby facilitating the generation of high-quality results.

A preliminary version of this work was presented in [10]. In this extended version, we provide a **systematic expansion** of the original study by comprehensively formulating the use of coherent correlations as an auxiliary supervision signal for speech-preserving emotion generation. Compared to the preliminary study, this work deepens the formulation and analysis from several critical perspectives. First, we extend the modeling of local correlations in facial animations to both spatial and temporal domains, enabling more expressive representations that lead to consistent improvements across different generative paradigms. Second, we investigate alternative formulations for capturing visual correlations, providing both theoretical insights and empirical comparisons of their effectiveness. Third, we introduce a correlation-aware adaptive weighting strategy to dynamically balance spatial and temporal correlation objectives according to their relative complexity. Finally, we conduct additional experiments to further validate the effectiveness of the proposed approach and demonstrate its generalization across diverse emotional talking head generation frameworks, including Transformer-based and Diffusion-based models.

The main contributions of this work can be summarized as follows:

- We propose a spatial-temporal coherent correlation learning (STCCL) framework, which captures spatial correlations between adjacent local regions within individual frames and temporal correlations across consecutive frames. As a supervision mechanism, it is orthogonal to existing model designs and can be seamlessly integrated into mainstream speech-preserving emotion generation pipelines as a plug-and-play enhancement.
- We introduce a correlation-aware adaptive weighting strategy that dynamically adjusts the contributions of spatial and temporal correlation objectives according to their relative complexity, facilitating more balanced and effective supervision.
- We investigate two representative formulations for modeling visual correlations, namely Visual Disparity and Correlation Matrix, and provide both analytical and empirical comparisons to illustrate their complementary strengths in capturing local details and global facial dynamics.
- We conduct extensive experiments by incorporating STCCL into representative SPFEM methods as well as recent emotional talking head generation frameworks (including Transformer-based and Diffusion-based models). The results consistently demonstrate the effectiveness and generalization capability of the proposed supervision strategy. The implementation details, including code and pretrained models, are publicly available at https://jianmanlincjx.github.io/STCCL/.

II. Related Work

In this section, we review the literature relevant to our work, categorized into three paradigms. First, we discuss general Facial Expression Manipulation (Section II-A), which focuses on

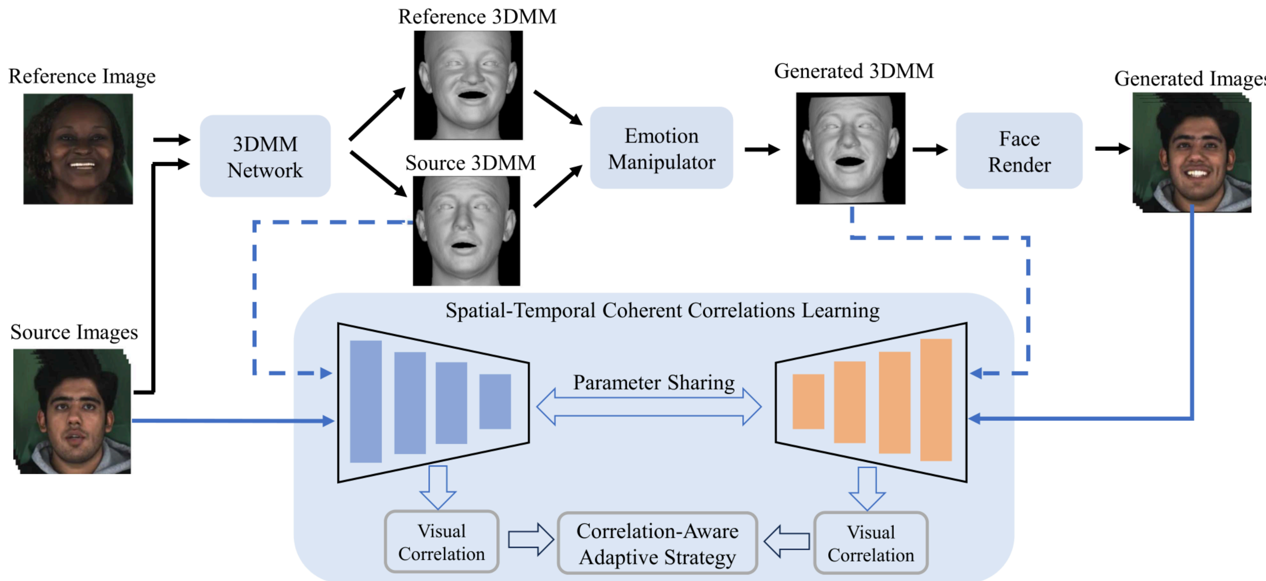

Fig. 4. Integration of STCCL into a standard Source → Intermediate → Rendering workflow (illustrated with NED [4] as a representative backbone). STCCL introduces an additional supervision signal that enforces spatial-temporal correlation consistency between the source input and generated outputs, applicable to both intermediate representations (e.g., 3DMM) and final rendered images. **Crucially, STCCL acts as a plug-and-play module that can be readily integrated into general SPFEM models without requiring paired supervision, effectively mitigating the long-standing conflict between vivid emotion manipulation and precise speech preservation.**

emotion editing but often neglects speech constraints. Second, we examine Emotional Talking Head Generation (Section II-B), covering recent disentanglement and diffusion-based approaches that synthesize videos from audio. Finally, we detail Speech-Preserving Facial Expression Manipulation (Section II-C), the specific task of this paper, and highlight the shared limitation across all paradigms: the inevitable interference of emotion-induced geometric deformations with speech-driven articulatory movements, particularly in the absence of paired supervision.

A. Facial Expression Manipulation

Early research leveraged general image-to-image translation frameworks [3], [11], [12], [13] or specialized conditional GANs [1], [14], [15], [16], [17], [18], [19] to alter facial attributes. For instance, ExprGAN [15] and GANmut [18] utilize conditional frameworks to control expression intensity or learn interpretable emotion spaces, while GANimation [20] employs Action Unit (AU) annotations [6] for fine-grained control over facial movements. To extend these capabilities to the video domain, methods like Head2Head++ [2] introduce sequential generators to enforce temporal consistency. However, a critical limitation persists across these generative approaches: they primarily prioritize visual expression transfer over speech preservation. Without explicit constraints, global expression changes often inadvertently corrupt lip synchronization, as the generated mouth shapes tend to drift towards training data biases rather than adhering to the spoken content.

More recently, StyleGAN-based approaches have gained prominence due to their high-quality disentangled latent spaces [5], [21]. These methods typically project input frames into latent codes via optimization [21], [22], [23] or encoders [24], [25], [26], [27], [28], [29], [30], followed by latent manipulation [31]. To ensure temporal coherence in videos, techniques like PTI [32] and STIT [33] propose fine-tuning generators on specific video clips, while TCSVE [34] incorporates explicit temporal consistency losses. RIGID [35] further advances this by learning inherent coherence for emotion-agnostic editing. Despite these advancements, StyleGAN-based methods face two significant hurdles: (1) Practicality and Efficiency: Most approaches are inherently video-specific and computationally expensive, requiring per-video fine-tuning or inversion. Moreover, identifying purely orthogonal editing directions that isolate emotion without affecting identity or background is notoriously difficult and labor-intensive. (2) Precision in Speech Preservation: Since these methods predominantly rely on global latent transformations, they often inadvertently alter speech-related regions (i.e., the mouth). Crucially, under unpaired training settings, they lack effective supervision to rigorously distinguish between emotion-induced deformations and articulatory movements, leading to a persistent trade-off between expression intensity and lip synchronization.

B. Emotional Talking Head Generation

Unlike general expression editing, emotional talking head generation synthesizes video portraits driven by audio and emotion. Initial efforts focused on **semantically disentangled representation learning**. Methods like EAMM [36], GC-AVT [37], and StyleTalk [7] employ latent codes or 3DMMs to explicitly separate emotion from speech and identity. Similarly, EDTalk [38] advances this by decomposing facial dynamics into distinct orthogonal latent spaces (i.e., mouth, pose, and expression) for independent control. Based on the premise that emotion semantics can be orthogonally combined with audio features, approaches like MEAD [39] and Emotion-AV [40] directly inject style labels, while EAT [8] achieves control via parameter-efficient adaptations on pre-trained Transformers. However, relying on implicit disentanglement often fails to handle large-scale geometric deformations, causing emotional expressions to interfere with fine-grained lip synchronization.

Recently, the field has shifted towards **diffusion-based generative frameworks** for their superior synthesis quality. General audio-driven methods like EMO [41] bypass intermediate 3D models to capture nuanced dynamics directly from audio, while DiffTalk [42] formulates the task within a Latent Diffusion Model (LDM) using landmark conditions. More recently, DICE-Talk [9] integrated explicit emotion conditioning into the probabilistic denoising process. Despite impressive texture fidelity, the inherent stochasticity of probabilistic generation poses a significant challenge for fine-grained control. While excelling in diversity, these models often struggle to maintain rigorous articulatory accuracy, leading to semantic ambiguity where generated mouth shapes deviate from the intended spoken content.

Critically, a fundamental bottleneck persists across both paradigms: the scarcity of paired data forces a reliance on self-driven training. Consequently, these methods lack explicit supervision to rigorously balance emotional expressiveness with articulatory precision. This highlights the necessity of our STCCL, which mines intrinsic correlations to construct a pseudo-paired supervisory signal, providing the deterministic guidance missing in current frameworks.

C. Speech-Preserving Facial Expression Manipulation

Distinct from generating videos from scratch (Section II-B) or optimizing for specific videos (Section II-A), SPFEM frameworks aim to be generalizable, modifying emotions in any source video while strictly preserving speech-related mouth animations. Early attempts, such as ICface [1] and Wav2Lip-Emotion [43], utilized interpretable control signals (e.g., AUs) or pre-trained emotion objectives to guide generation. However, these 2D-based methods often struggle to maintain identity consistency and suffer from limited visual fidelity in the generated results.

3D Morphable Models (3DMMs) [17], [44], [45], [46], [47] provide an explicit representation for modeling facial movements. In particular, [48] demonstrate that 3DMMs are capable of capturing large-scale deformations, such as wide mouth openings in anger or raised eyebrows in joy, which significantly influence the perceived emotional valence of an expression. These capabilities make 3DMMs particularly well-suited for integration into the SPFEM model. Several works have leveraged 3DMMs for semantic facial expression manipulation. For instance, DSM [49] enables semantic video editing through neural rendering combined with 3DMMs, providing intuitive control over facial expressions and introducing an AI tool that maps semantic labels into the Valence-Arousal space, which are then translated into photorealistic 3D facial expressions. Similarly, NED [4] proposes a framework based on a parametric 3D face representation that disentangles facial identity from head pose and expressions. By employing deep domain translation and neural face rendering, NED achieves consistent and photorealistic facial expression manipulation. However, a key limitation of traditional 3DMM-based methods is their inability to capture fine-grained details such as subtle wrinkles or color variations. To address this, [48] propose a novel perspective, treating the task as a special case of motion information editing: they use 3DMMs to capture large-scale facial movements while modeling detailed appearance features with a StyleGAN-based texture map.

Despite these architectural advancements, a fundamental bottleneck remains across existing paradigms: the reliance on unpaired training data. The absence of paired supervision (i.e., same speech content with different emotions) often leads to suboptimal trade-offs between emotion manipulation and speech preservation. Addressing this gap, we move beyond architectural modifications and propose a novel supervisory signal. By modeling the inherent spatial-temporal coherent correlations between the source and generated video sequences, we construct a robust pseudo-paired supervision, thereby enhancing both the quality of emotion editing and the precision of lip synchronization.

III. Visual Correlation Analysis

This section provides empirical evidence for our core assumption: local facial animations exhibit strong correlations when a speaker articulates identical content across different emotions. To quantify this, we define visual correlation using visual disparities and correlation matrices, as capturing local motion disparities and structural dependencies is essential for realistic facial animation. We then apply these metrics to analyze spatio-temporal relationships in paired data. Specifically, correlations between corresponding adjacent regions across different emotional states are defined as positive samples, while those between non-corresponding regions serve as negative samples. Our statistical analysis reveals consistently high correlation values for positive samples compared to near-zero values for negative ones, thereby empirically validating our hypothesis.

A. Visual Correlation Definition

To model the consistent local facial animations across different emotional expressions, defining visual correlation—how adjacent facial regions within images relate to each other—is crucial. We introduce two mechanisms: visual disparities and correlation matrices, both designed to evaluate the visual correlation between any two regions i and j in an image feature I^f, spatially and temporally.

Visual Disparities: In the study by CCPL [50], it was demonstrated that for the Image-to-Image transition task, aligning the pixel-wise differences between adjacent local regions of the input image and the corresponding regions in the generated image can enhance the quality of the generated images. This insight suggests that focusing on the local pixel-level correlations between corresponding regions of the input and output images can lead to better alignment. Building on this, we define visual disparities as a way to formally represent these local visual correlations. Since visual disparities focus on the immediate pixel-wise differences between specified regions, the visual correlation between regions i and j can be calculated using:

$$V(I^f, i, j) = I^f(i) - I^f(j) \tag{1}$$

where $I^f(i)$ and $I^f(j)$ represent the pixel intensities at regions i and j, respectively. This method is particularly useful for capturing subtle variations in expression, essential for high-frequency detail analysis.

Correlation Matrix: Visual disparity directly focuses on the information differences between adjacent regions within an image, making it suitable for image-to-image tasks. Inspired by the use of correlation matrices in [51], [52] for capturing complex relationships in various domains, we introduce the correlation matrix as an alternative visual correlation mechanism for emotion transformation tasks. The correlation matrix begins by constructing a correlation matrix M through the matrix multiplication of the feature vectors and their transposes:

$$M = F \cdot F^T \tag{2}$$

where F is a matrix with columns representing feature vectors from the regions. The visual correlation between regions i and j in I^f is obtained from M as follows:

$$C(I^f, i, j) = M_{ij} \tag{3}$$

where M_{ij} quantifies the correlation between the features from regions i and j. Correlation matrices capture complex relationships and dependencies between different facial regions, offering a holistic view of facial dynamics that is crucial for understanding how changes in one region affect others during emotional expressions.

Unified Representation: To facilitate a generalized discussion, we define a unified function $Z(\cdot)$, which represents either V or C based on the selected mechanism. To distinguish between spatial and temporal dimensions, we use specific notations that explicitly define the input sources and the domain of the indices:

- *Spatial Correlation (s):* When indices i and j denote different **spatial locations** within a single image frame I^f, the spatial correlation is defined as:

$$I^{fs}_{ij} = Z(I^f, i, j). \tag{4}$$

- *Temporal Correlation (t):* When i and j denote different **temporal indices** (i.e., frame counts) within a multi-scale feature sequence $\{I^{f1}, I^{f2}, \ldots, I^{fT}\}$ at a fixed spatial location, the temporal correlation is defined as:

$$I^{ft}_{ij} = Z(\{I^{f1}, \ldots, I^{fT}\}, i, j). \tag{5}$$

This dual-notational framework explicitly differentiates intra-frame spatial consistency (where i, j index spatial positions in I^f) from inter-frame temporal coherence (where i, j index temporal frames across the sequence $\{I^{f1}, \ldots, I^{fT}\}$).

B. Spatial Correlation Mining

Facial movement patterns intuitively maintain consistency when a speaker articulates identical content across different emotions. We first explore this consistency in the spatial domain. Formally, given two images x and y of the same speaker expressing the same content with two different emotions, we extract their features x^f and y^f using a pre-trained network [53]. We then evaluate the similarity between the spatial correlations of corresponding and non-corresponding regions using the cosine similarity function φ:

$$\begin{aligned} s^p_{ij} &= \varphi(x^{fs}_{ij}, y^{fs}_{ij}) \\ s^n_{ik} &= \varphi(x^{fs}_{ij}, y^{fs}_{ik,k\neq j}) \end{aligned} \tag{6}$$

where x^{fs}_{ij} and y^{fs}_{ij} denote the spatial correlations between regions i and j as defined in Section III.

By analyzing thousands of region pairs across neutral-to-emotion transitions, we observe a stark contrast in similarity distributions. As illustrated in Fig. 3(a), while the average similarities for non-corresponding regions (s^n) approach zero, those for corresponding regions (s^p) range significantly from 0.43 to 0.49. Further validation using correlation matrices yields a consistent phenomenon, as shown in Fig. 3(b). These results confirm that strong spatial correlations exist between corresponding local regions across different emotional states, provided the spoken content remains identical. Leveraging this insight, we can integrate these correlations as a supervisory signal to enhance SPFEM performance. Such an integration would specifically aim to enforce intra-frame visual consistency between the local regions of the input image and their corresponding counterparts in the generated output.

C. Temporal Correlation Mining

Extending the spatial analysis, we anticipate even more pronounced consistency in the temporal dimension, driven by the evolving dynamics of facial expressions as a speaker conveys identical content across varying emotions. Moving beyond static images, we consider continuous image sequences denoted as X and Y, with their respective features represented as X^f and Y^f. To quantify visual correlation in the temporal dimension, we fix the spatial coordinates and extract features across the temporal axis. The similarities between temporal correlations for corresponding and non-corresponding regions are formulated as:

$$\begin{aligned} s^p_{ij} &= \varphi(X^{ft}_{ij}, Y^{ft}_{ij}) \\ s^n_{ik} &= \varphi(X^{ft}_{ij}, Y^{ft}_{ik,k\neq j}) \end{aligned} \tag{7}$$

where X^{ft}_{ij} and Y^{ft}_{ij} denote the visual correlations of specific regions across adjacent frames i and j.

As illustrated in Fig. 3(c) and (d), the results reveal that temporal correlations between corresponding regions are significantly stronger than their spatial counterparts. Given that X and Y articulate the same content over time, the temporal dimension more faithfully reflects the articulatory consistency than the spatial domain. Consequently, further investigating these temporal coherent correlations and integrating them into the SPFEM model could provide a superior supervisory signal. This integration is expected to enhance inter-frame visual consistency, effectively aligning the motion trajectories of the generated video with the source input.

Our experiments confirm a pronounced consistency in visual correlations across different emotional states when the spoken content remains identical. Specifically, the intra-frame correlations between adjacent local regions in one emotional expression closely match their counterparts in others. This consistency extends to the temporal domain, where inter-frame correlations between successive frames in one emotional sequence mirror those in equivalent regions of a different emotion. These findings motivate us to leverage such inherent spatio-temporal correlations as a direct supervisory signal to enhance the training of the SPFEM model.

IV. ST-CCL

The proposed ST-CCL framework initially establishes a spatially coherent correlation metric to align the visual dependencies of adjacent local regions between the source input and the generated output. Concurrently, it develops a temporal coherent correlation metric to ensure that cross-frame visual correlations within specific facial regions remain consistent across the source and generated sequences. To address the varying complexity of facial dynamics, we introduce a correlation-aware adaptive strategy that dynamically assigns higher importance to challenging regions while modulating weights for simpler areas. As a versatile, plug-and-play supervision signal, ST-CCL can be seamlessly integrated into various generation stages, including intermediate geometric representations (e.g., 3DMM [54]) and final rendered images. To visualize this integration within a mainstream generative paradigm, we illustrate the overall pipeline in Fig. 4, using the two-stage NED [4] framework as a representative backbone to demonstrate ST-CCL's deployment across both intermediate and final synthesis stages.

A. Spatial Coherent Correlation Metric Learning

As established in Section III, strong inherent correlations exist between adjacent facial regions when a speaker expresses identical content across different emotions. We further strengthen these correlations by learning a *Spatial Coherent Correlation* (SCC) metric using paired data. This metric is designed to assign higher values to correlations between corresponding adjacent regions while suppressing non-corresponding counterparts, thereby providing high-fidelity supervisory signals for the generation process.

Formally, given two images of the same speaker articulating identical content with different emotions, we must extract corresponding regions i and j. However, head pose misalignment typically exists between such image pairs, which can compromise the accuracy of correlation calculations. To address this, we introduce a pose alignment module. We first extract facial landmark coordinates using a pre-trained OpenCV-based model for both images and utilize these coordinates to compute an affine transformation matrix. This matrix is subsequently applied to align the images, yielding a pair (x, y) that ensures precise geometric correspondence.

The details of the SCC metric are illustrated in the upper panels of Fig. 5. Due to the inherent locality and translation invariance of Convolutional Neural Networks (CNNs) [53], [55], spatial locations within the feature maps naturally correspond to localized regions in the original image. As the network depth increases, the receptive field expands accordingly. To capture multi-level visual dependencies, we employ the output of multiple convolutional layers to obtain multi-scale features x^f and y^f:

$$x^f = \{\varepsilon_l(x)\}_{l=1}^{L}$$
$$y^f = \{\varepsilon_l(y)\}_{l=1}^{L} \tag{8}$$

where $\varepsilon_l(\cdot)$ denotes the feature extractor at layer l, and L is the total number of layers. We adopt the Arcface network [53] and utilize the output of four convolutional layers ($l \in [1, 2, 3, 4]$) where the spatial dimensions decrease progressively. For any two regions i and j, the visual correlation is first computed via the predefined function $Z(\cdot, i, j)$ as defined in Section III-A and then projected by a mapping function $f(\cdot)$:

$$x_{ij}^{fs} = f(Z(x^f, i, j))$$
$$y_{ij}^{fs} = f(Z(y^f, i, j)), \tag{9}$$

where $f(\cdot)$ is implemented as two stacked fully-connected layers with ReLU activation. Note that the same mapping function $f(\cdot)$ is applied to both source and target features to ensure they are projected into a common latent space for alignment. The mapping function projects the correlations into a shared feature space to facilitate effective correlation modeling.

Inspired by recent advancements in self-supervised learning [50], [56], [57], [58], we define a contrastive loss to train the SCC metric:

$$\ell_{xy}^{ij,s} = -\log \frac{\exp(x_{ij}^{fs} \cdot y_{ij}^{fs}/\tau)}{\exp(x_{ij}^{fs} \cdot y_{ij}^{fs}/\tau) + \sum_{k=1,k\neq j}^{m} \exp(x_{ij}^{fs} \cdot y_{ik}^{fs}/\tau)}, \tag{10}$$

where τ is a temperature hyper-parameter set to 0.07. The final SCC loss is defined as the summation over all image pairs and corresponding region pairs:

$$\mathcal{L}_{sccl} = \sum_{i,j} \sum_{x,y} \ell_{xy}^{ij,s}. \tag{11}$$

By minimizing $\mathcal{L}_{sccl}$ through backpropagation, the SCC metric learns to identify and exploit inherent spatial coherence within the paired data. During SPFEM model training, this pre-trained SCC metric imposes spatial consistency constraints on the visual correlations between corresponding local regions of the generated image and the source input, ensuring superior spatial alignment and realistic facial dynamics.

B. Temporal Coherent Correlation Metric Learning

While the SCC metric ensures intra-frame spatial alignment, preserving inter-frame temporal coherence is equally vital for realistic facial animation. As analyzed in Section III, when a speaker articulates the same content across different emotions, the visual correlations within specific regions across consecutive

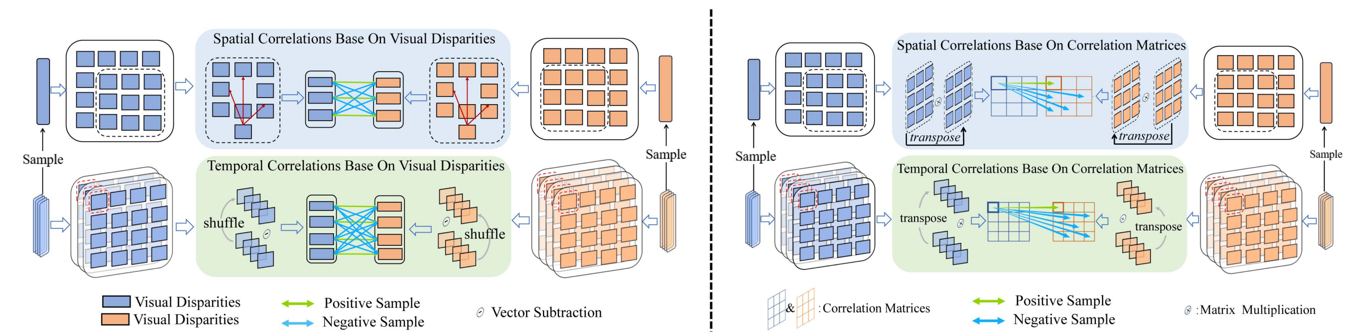

Fig. 5. **Left half:** Illustration of the Spatial-Temporal Coherent Correlation Learning (STCCL) based on **Visual Disparity**. It constructs dense contrastive pairs within the feature space, designating corresponding spatially adjacent and temporally consecutive regions of the source and generated sequences as positive samples, while treating non-corresponding counterparts as negative samples. **Right half:** The algorithmic variant based on **Correlation Matrices**. Unlike visual disparity, it captures structural dependencies by computing correlation maps via matrix multiplication, providing a holistic view of facial dynamics.

frames exhibit high similarity. By learning a *Temporal Coherent Correlation* (TCC) metric from paired data, we can effectively align the dynamic visual trajectories of the source input sequence with those of the generated output.

The architecture of the TCC metric is illustrated in the lower panels of Fig. 5. Formally, consider two sequences $X = \{x^1, x^2, \ldots, x^T\}$ and $Y = \{y^1, y^2, \ldots, y^T\}$ from the same speaker, representing identical content with different emotions. Their corresponding multi-scale feature sequences are denoted as $X^f = \{x^{f1}, x^{f2}, \ldots, x^{fT}\}$ and $Y^f = \{y^{f1}, y^{f2}, \ldots, y^{fT}\}$. To compute the temporal correlation, we fix a specific spatial region and extract its features along the temporal axis. For any two temporal indices i and j (representing frames within a temporal span), the visual correlation is computed by applying the function Z to the feature sequence, followed by the mapping function $f(\cdot)$:

$$\begin{aligned} X_{ij}^{ft} &= f(Z(X^f, i, j)) \\ Y_{ij}^{ft} &= f(Z(Y^f, i, j)), \end{aligned} \tag{12}$$

where X_{ij}^{ft} and Y_{ij}^{ft} encapsulate the temporal dynamics between frames i and j. Similar to the spatial metric, $f(\cdot)$ projects these temporal correlations into a shared latent space to facilitate effective comparison. Here, the same mapping function is applied to both the source and generated sequences to ensure consistent feature alignment.

We employ contrastive learning to strengthen the correlation of positive temporal pairs while suppressing non-corresponding ones:

$$\ell_{xy}^{ij,t} = -\log \frac{\exp(X_{ij}^{ft} \cdot Y_{ij}^{ft}/\tau)}{\exp(X_{ij}^{ft} \cdot Y_{ij}^{ft}/\tau) + \sum_{k=1,k\neq j}^{m} \exp(X_{ij}^{ft} \cdot Y_{ik}^{ft}/\tau)}, \tag{13}$$

where τ is the temperature hyper-parameter. The total TCC loss is the summation over all sequence and temporal region pairs:

$$\mathcal{L}_{tccl} = \sum_{i,j} \sum_{X,Y} \ell_{xy}^{ij,t}. \tag{14}$$

By minimizing $\mathcal{L}_{tccl}$ through backpropagation, the TCC metric learns to identify and exploit inherent temporal coherence within the paired data. During SPFEM model training, the TCC metric enforces temporal consistency constraints on the generated sequences. By aligning the temporal correlation information between specific regions of the input and generated sequences, this method ensures more stable emotion manipulation and consistently improves the synchronization between lip movements and audio content.

C. Correlation-Aware Adaptive Strategy

Once the SCC metric and TCC metric are learned, we can use $\mathcal{L}_{sccl}$ and $\mathcal{L}_{tccl}$ between the input source (source 3DMM) and output generated (generated 3DMM) images as additional supervision. Here, we observe a notable phenomenon that the complexity varies across different facial regions. For example, it is more complex and challenging for the mouth regions as it change dramatically when the speaker is talking. In contrast, when the regions are positioned at the cheek area, the visual correlation is reduced, promoting a faster convergence. This underscores the region-specific sensitivity of visual correlation. Consequently, by enabling the SCC metric and TCC metric to independently discern and dynamically modulate learning strategies for specific regions, we can adaptively enhance the extraction of spatial-temporal coherent correlation information.

Inspired by the idea of [59], CAAS is engineered to enable the SCC metric and TCC metric to discern this variability and tailor its learning strategies based on the characteristics of the regions, prioritizing those with higher learning complexity. Specifically, when the visual correlation $x_{ij}^{f,s}$ or $x_{ij}^{f,t}$ is larger, this region is deemed a challenging region, to which we assign a larger weight value. Conversely, when $x_{ij}^{f,s}$ or $x_{ij}^{f,t}$ is small, it is considered a simple region that can achieve rapid convergence; we therefore reduce the weight for those simple regions. To this end, we propose to assign different weights according to the visual correlation, formulated as:

$$\begin{aligned} w_{ij}^{s} &= \lambda \cdot \mathrm{sigmoid}(x_{ij}^{fs})^{r} \\ w_{ij}^{t} &= \lambda \cdot \mathrm{sigmoid}(x_{ij}^{ft})^{r} \end{aligned} \tag{15}$$

where w_{ij}^{s} represents the weight value of adjacent regions i and j for images x and y. λ and r are hyper-parameters, both of which are set to 2 to ensure a reasonable weight. The final loss function can be defined as:

$$\mathcal{L}_{stccl} = \sum_{i,j}\sum_{x,y} w_{ij}^{s} \cdot \ell_{xy}^{ij,s} + \sum_{i,j}\sum_{x,y} w_{ij}^{t} \cdot \ell_{xy}^{ij,t} \tag{16}$$

Current generation paradigms generally fall into two categories: multi-stage frameworks that leverage intermediate representations (e.g., 3DMM parameters [4]) and direct synthesis models (e.g., GANs [1] or diffusion [9]). Consequently, "visual information" in our context refers to either intermediate geometric features or final rendered images. As shown in Fig. 4, $\mathcal{L}_{stccl}$ is architecture-agnostic and universally applicable to both streams. *Detailed implementations, including integrations with NED [4], ICface [1], EAT [8], and DICE-Talk [9], are provided in the supplementary materials due to page limits.*

V. Experiments

A. Experimental Setup

We performed experiments on the MEAD dataset [39], which contains 60 speakers, and each speaker records 30 videos in each emotional state (i.e., neutral, happy, angry, surprised, fear, sad, and disgusted). Here, we selected videos of 36 speakers that have 7,560 videos to train the STCCL algorithm. To evaluate the SPFEM model's performance, we select 6 non-overlapped speakers (M003, M009, W029, M012, M030, and W015) that have 1,260 videos. We randomly selected 90% as the training set and the rest 10% as the test set similar to previous works [4]. We additionally employ the STCCL algorithm on the well-established RAVDESS dataset [60] without the need for re-training. Specifically, we focus on 6 speakers (actors 1-6) encompassing 168 videos. Similarly, 90% of the videos are randomly chosen for the training set, while the remaining 10% constitute the test set.

B. Evaluation Protocol

In this work, we use these metrics for evaluation: 1) Frechet Arcface Distance (FAD) gauges video realism by comparing feature vectors of generated and real videos using advanced face recognition technology [53]. Lower FAD values indicate better realism. 2) Cosine Similarity (CSIM) assesses emotional similarity between generated and target emotional videos using a state-of-the-art expression recognition network, with higher CSIM values denoting greater similarity. 3) Lip Sync Error Distance (LSE-D) [61] evaluates lip-audio accuracy using a pre-trained model [62] to measure the disparities between lip and audio representations, with smaller LSE-D indicating a higher lip-audio accuracy. We present the results of two settings: Intra-ID, where the emotion reference and source video share the same speaker, and Cross-ID, involving different speakers.

C. Comparison With Baseline Methods

To rigorously evaluate the effectiveness and versatility of our STCCL algorithm, we conduct a comprehensive evaluation strategy. First, we integrate STCCL (both Ours_VD and Ours_CM) into two foundational SPFEM models—the **one-stage ICface** [1] and the **two-stage NED** [4]—to verify its core efficacy across different architectural pipelines. Second, to demonstrate that STCCL serves as a universal supervision signal, we extend its application to the broader field of emotion-aware talking head generation. By incorporating STCCL into the transformer-based **EAT** [8] and the diffusion-based **DICE-Talk** [9], we validate its **architecture-agnostic efficacy**, showcasing consistent performance gains across both mainstream end-to-end pipelines and emerging probabilistic paradigms. The baseline methods are summarized as follows:

- *NED [4]:* A two-stage framework that blends 3DMM parameters of the source identity and target emotion to achieve expression manipulation.
- *ICface [1]:* A one-stage method that employs Action Units (AUs) to depict facial expressions and transition the source face to the target emotion.
- *EAT [8]:* An end-to-end transformer-based architecture that leverages emotional adaptation modules atop a pre-trained emotion-agnostic talking head transformer.
- *DICE-Talk [9]:* A cutting-edge diffusion-based paradigm that embeds audio and emotion priors into a probabilistic denoising process to achieve high-fidelity generation.

Notably, since EAT and DICE-Talk utilize predetermined emotional guidance rather than reference-based extraction, they do not inherently distinguish between Intra-ID and Cross-ID settings. To maintain consistency, we report the performance of these STCCL-enhanced models under the cross-identity setting for a fair comparison with other baselines.

1) Quantitative Comparisons: We first present the performance comparisons on MEAD in Table I, which reports the averaged results across all seven emotions (detailed per-emotion metrics are provided in the Supplementary Material). Integrating STCCL into NED consistently improves FAD, LSE-D, and CSIM. Specifically, in the Cross-ID setting, Ours_VD outperforms the baseline by reducing FAD from 4.448 to 4.169 and LSE-D from 9.906 to 9.216. The substantial LSE-D gain validates that our supervision accentuates intrinsic spatial-temporal correlations, ensuring consistent mouth shape preservation. Similar gains are observed with ICface (e.g., Intra-ID FAD 6.795→6.634), demonstrating adaptability to subject migration. Moreover, Ours_CM consistently enhances NED (e.g., Cross-ID CSIM reaching 0.799), proving robustness across different correlation mechanisms. Notably, STCCL consistently outperforms its preliminary version (ASCCL); for instance, Ours_VD improves NED (ASCCL) Intra-ID FAD from 1.234 to 1.077, confirming that the comprehensive spatial-temporal modeling and refined algorithms in this extended version yield more robust supervision.

To demonstrate that STCCL is an architecture-agnostic and future-proofed supervision signal, we extend its application to state-of-the-art emotional talking head generation models: the transformer-based EAT [8] and the diffusion-based DICE-Talk [9]. As reported in Table I, STCCL consistently enhances these advanced generative frameworks. For EAT, Ours_VD reduces the LSE-D from 10.023 to 9.381, effectively alleviating the common trade-off between emotional expressiveness and

TABLE I

QUANTITATIVE COMPARISON OF FAD, LSE-D, AND CSIM BETWEEN OUR FRAMEWORK AND COMPETING METHODS ON THE MEAD AND RAVDESS DATASETS. RESULTS ARE REPORTED UNDER BOTH INTRA-IDENTITY AND CROSS-IDENTITY SETTINGS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINE, RESPECTIVELY. THE "ASCCL" ROWS REFER TO THE RESULTS FROM OUR CONFERENCE VERSION [10]. DETAILED METRICS FOR INDIVIDUAL EMOTIONS ARE PROVIDED IN THE SUPPLEMENTARY MATERIAL TO MAINTAIN A FOCUSED PRESENTATION OF THE MAIN RESULTS.

Methods	MEAD Dataset						RAVDESS Dataset					
	Intra-ID			Cross-ID			Intra-ID			Cross-ID		
	FAD↓	LSE-D↓	CSIM↑	FAD↓	LSE-D↓	CSIM↑	FAD↓	LSE-D↓	CSIM↑	FAD↓	LSE-D↓	CSIM↑
Backbone: ICface [1] (One-stage GAN-based)												
Baseline	6.795	10.083	0.775	9.540	11.238	0.688	8.443	8.480	0.755	9.424	11.539	0.677
ASCCL [10]	6.672	9.431	0.801	9.480	10.375	0.726	7.301	8.044	0.765	9.086	10.313	0.682
Ours_VD	**6.634**	**9.411**	0.806	**9.445**	**10.311**	0.730	**7.289**	**8.027**	0.772	**8.900**	10.315	0.687
Ours_CM	6.651	9.412	**0.810**	9.460	10.312	**0.732**	7.294	**8.027**	**0.775**	8.902	**10.312**	**0.691**
Backbone: NED [4] (Two-stage 3DMM-based)												
Baseline	2.108	9.454	0.831	4.448	9.906	0.773	3.057	7.562	0.825	5.412	8.034	0.760
ASCCL [10]	1.234	9.340	0.900	4.264	9.238	0.791	3.039	7.450	0.830	4.808	7.820	0.761
Ours_VD	**1.077**	9.335	0.914	**4.169**	**9.216**	0.795	**2.938**	**7.404**	0.835	**4.774**	**7.803**	0.767
Ours_CM	1.103	**9.330**	**0.916**	4.253	9.225	**0.799**	2.996	**7.404**	**0.839**	4.794	7.804	**0.772**
Backbone: EAT [8] (Transformer-based)												
Baseline	-	-	-	7.464	10.023	0.706	-	-	-	5.889	7.921	0.700
ASCCL [10]	-	-	-	7.350	9.750	0.718	-	-	-	5.541	7.682	0.716
Ours_VD	-	-	-	**7.198**	**9.381**	0.733	-	-	-	**5.189**	**7.440**	0.732
Ours_CM	-	-	-	7.256	9.389	**0.740**	-	-	-	5.260	7.449	**0.740**
Backbone: DICE-Talk [9] (Diffusion-based)												
Baseline	-	-	-	2.666	9.082	0.777	-	-	-	3.241	7.778	0.775
ASCCL [10]	-	-	-	2.590	8.980	0.785	-	-	-	3.106	7.638	0.785
Ours_VD	-	-	-	**2.470**	**8.872**	0.797	-	-	-	**2.973**	**7.497**	0.795
Ours_CM	-	-	-	2.532	8.879	**0.803**	-	-	-	3.034	7.499	**0.801**

lip-sync accuracy. This quantitatively validates our hypothesis that modeling spatial-temporal correlations stabilizes the mouth articulations against the jitter introduced by large-scale emotional deformations. Even for the diffusion-based DICE-Talk, which already exhibits a high-fidelity baseline, STCCL further pushes the boundaries of realism and synchronization, reducing FAD from 2.666 to 2.470 and LSE-D from 9.082 to 8.872. These results prove that STCCL acts as a universal robust supervision that can seamlessly complement both deterministic disentanglement-based and probabilistic generative paradigms, steering them toward superior optimization of fine-grained facial dynamics.

In multiple experiments, we found that the STCCL algorithm based on visual disparity performs better in terms of the FAD metric compared to the STCCL algorithm based on the correlation matrix. One possible reason is that the STCCL algorithm based on visual disparity aligns the pixel differences of adjacent regions in the input images of the SPFEM model with the pixel differences of the corresponding adjacent regions in the output images of SPFEM model in both the temporal and spatial dimensions. This pixel-level supervision can make the quality of the generated images more similar to the input images. At the same time, we found that the STCCL algorithm based on the correlation matrix performs better in terms of the CSIM metric compared to the STCCL algorithm based on visual disparity. We empirically believe that this is because correlation matrices capture complex relationships and dependencies between different facial regions, offering a holistic view of facial dynamics that is crucial for understanding how changes in one region affect others during emotional expressions, thus leading to better performance in the CSIM metric.

To rigorously validate the generalization capability of our approach, Table I reports the performance on the unseen RAVDESS dataset. Crucially, the STCCL module used in this experiment was trained exclusively on the MEAD dataset and subsequently integrated into the backbone models as a plug-and-play supervision signal without any fine-tuning. Under this challenging cross-domain setting, STCCL yields consistent improvements across diverse architectures. For ICface, Ours_VD decreases Intra-ID FAD and LSE-D by 1.154 and 0.453, respectively. In the Cross-ID setting, it reduces FAD and LSE-D by 0.524 and 1.224. For NED, Ours_CM achieves superior emotional similarity (Cross-ID CSIM 0.772vs. baseline 0.760). Remarkable generalization is also seen in talking head models: Ours_VD improves EAT (Cross-ID FAD 5.889→5.189) and DICE-Talk (reducing FAD to 2.973). These consistent gains across unseen domains and architectures confirm that STCCL learns a robust, universal representation of spatial-temporal correlations.

2) *Qualitative Comparisons:* In this section, we exhibit visualization results of the baseline NED methodology, both with and without the STCCL algorithm, on the MEAD and RAVDESS datasets, as illustrated in Fig. 6. Due to space constraints, qualitative comparisons for other baseline methods (e.g., ICface,

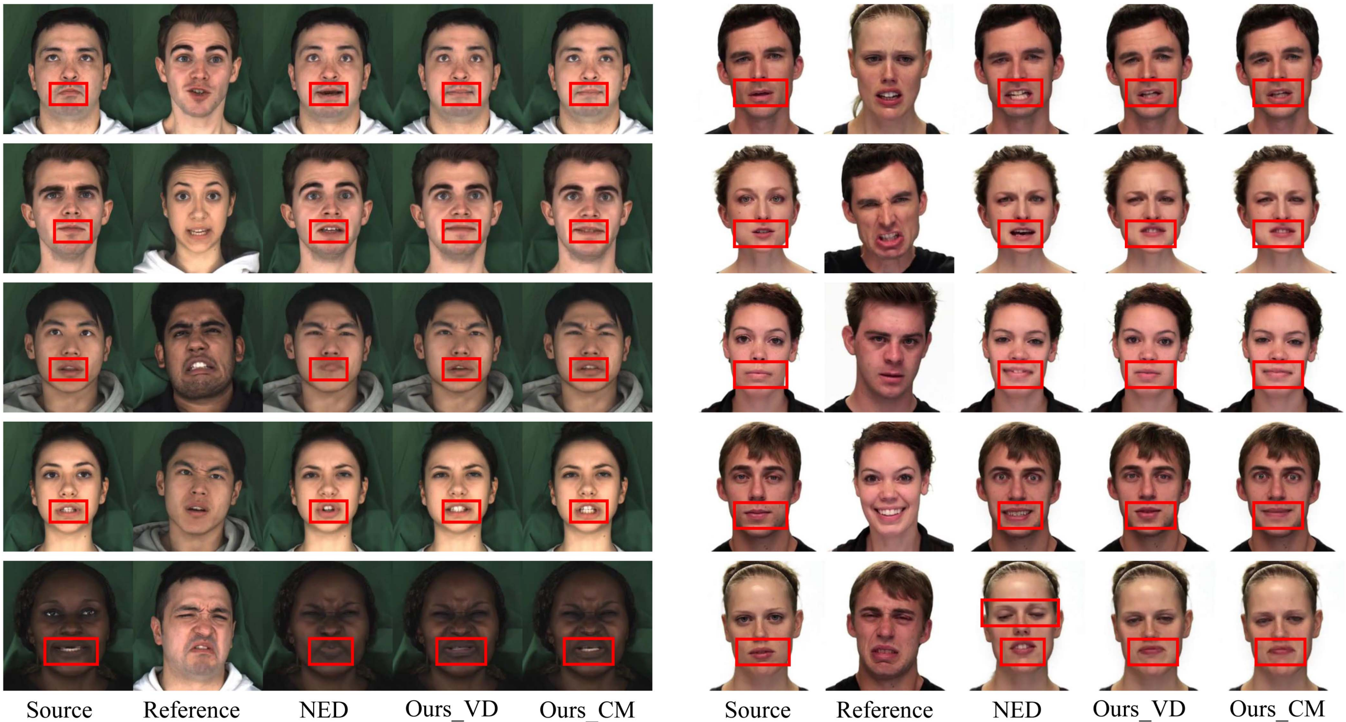

Fig. 6. Qualitative comparisons of NED with and without the proposed STCCL algorithm. **Left half:** The samples are selected from the MEAD dataset. **Right half:** The samples are selected from the RAVDESS dataset.

EAT, and DICE-Talk) are provided in the supplementary material. Consistent with the quantitative metrics, we dissect the qualitative performance from three key dimensions.

1) Realism: A common issue with the standalone NED [4] is the tendency for the eye region to appear overly closed, as exemplified in the third row/third column of the left half and the fifth row/third column of the right half of Fig. 6. Furthermore, the mouth region often suffers from distortion due to imprecise shape prediction. STCCL ameliorates these shortcomings by aligning the visual correlation consistency between inputs and outputs. As demonstrated in the fourth and fifth columns (Ours_VD and Ours_CM), our method ensures more natural eye states and precise mouth structures.

2) Emotional Similarity: Existing manipulation approaches often fail to accurately communicate the reference subject's emotions, frequently opting for a direct replication of facial components into the source (see the third column, fifth row of the right half of Fig. 6). By maximizing the alignment of spatial-temporal correlations, STCCL directs the model to prioritize the extraction of emotional semantics rather than merely duplicating pixels. Consequently, the NED methodology becomes more proficient in preserving the source's original contours while achieving high-fidelity emotional transference.

3) Lip-Audio Preserving Accuracy: The proposed CAAS is specifically designed to focus on the mouth area, which undergoes the most significant dynamic changes. STCCL maintains mouth shape consistency by constraining the visual correlations within these critical regions in positive samples. The results in the fourth and fifth columns demonstrate a superior ability to retain the source's articulatory precision compared to the NED baseline. For a more comprehensive evaluation, we provide additional video comparisons and qualitative results for EAT and DICE-Talk in the supplementary materials.

TABLE II
REALISM, EMOTION SIMILARITY, AND MOUTH SHAPE SIMILARITY RATINGS OF THE USER STUDY ON NED AND OUR STCCL ON MEAD DATASET

Emotion	Realism		Emotion similarity		Mouth shape similarity	
	NED	STCCL	NED	STCCL	NED	STCCL
Neutral	28%	72%	24%	76%	31%	69%
Angry	28%	72%	36%	64%	30%	70%
Disgusted	35%	65%	34%	66%	26%	74%
Fear	28%	72%	35%	65%	29%	71%
Happy	28%	72%	33%	67%	28%	72%
Sad	35%	65%	28%	72%	25%	75%
Surprised	30%	70%	27%	73%	24%	76%
Avg.	30%	70%	31%	69%	27%	73%

3) User Study: We conducted web-based user studies to compare the performance of NED with and without the STCCL (base on visual disparity) algorithm on the MEAD dataset. The study comprises three segments corresponding to the previously mentioned metrics: realism, emotion similarity with the reference emotion, and mouth shape similarity with the source video, covering seven basic emotions. For each emotion, we carefully selected 10 videos for both Intra-ID and Cross-ID settings, totaling 140 videos. Involving 25 participants, each participant was tasked with assessing the three aspects of each video. As detailed in Table II, the inclusion of the STCCL algorithm consistently outshines the baseline NED method across all seven emotions in all three metrics. On average, the integration of the STCCL

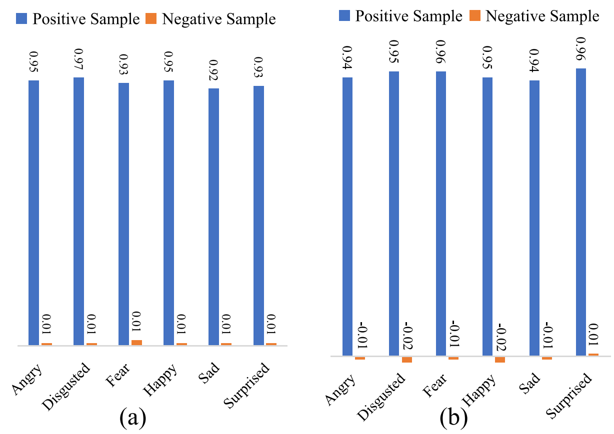

Fig. 7. (a) Average similarities of the positive and negative samples based on visual disparities and (b) Average similarities of the positive and negative samples based on correlation matrices, both after training the SCC metric in spatial space (compared to Fig. 3).

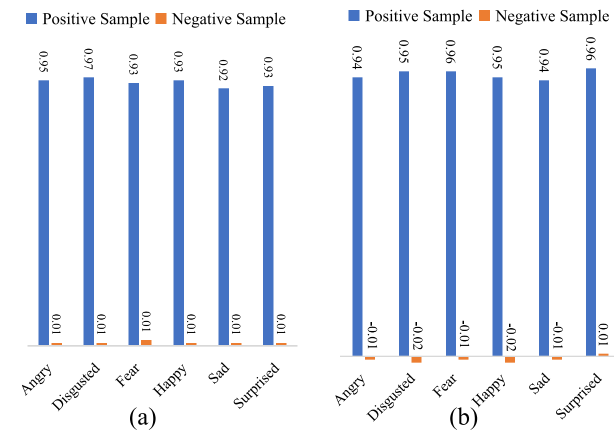

Fig. 8. (a) Average similarities of the positive and negative samples based on visual disparities and (b) Average similarities of the positive and negative samples based on correlation matrices, both after training the TCC metric in temporal space (compare to Fig. 3).

algorithm demonstrates consistent improvement, achieving a 40% higher rating in realism, a 38% higher rating in emotion similarity, and an impressive 46% higher rating in mouth shape similarity compared to the NED baseline. *Supplementary materials include user studies utilizing the ICface baseline on MEAD and employing both NED and ICface baselines on RAVDESS.*

D. Ablation Study

In this section, we conduct ablation experiments to analyze the role of each component in the STCCL algorithm. For a clear demonstration, we perform ablation experiments using the visual disparity-based STCCL on the ICface and NED methods on the MEAD dataset.

1) Analyses of STCCL Metric Learning: Our statistical analysis demonstrates an inherent spatial-temporal coherent correlation when a speaker expresses the same content across different emotions. Even untrained, the STCCL can approximate this correlation, as depicted in Fig. 3. By pre-training STCCL with paired data, we enhance the STCCL's ability to discern inherent visual correlation. As shown in Figs. 7 and 8, there is a notable improvement in the similarity for positive samples. This indicates that through spatial-temporal coherent correlation learning, we can further uncover the latent associations between data, providing a more efficient supervision signal for the training of the SPFEM model.

To verify the impact of performing STCCL metric learning on the training of the SPFEM model, we employ both trained and untrained STCCL to guide the SPFEM model's training, as depicted in Table III. When integrating STCCL into NED, even without utilizing paired data for training STCCL, it can also serve as guidance. In the Cross-ID setting, Ours untrained shows relative improvements over the NED baseline in the FAD, LSE-D, and CSIM metrics by 0.147, 0.463, and 0.012, respectively. The pre-trained STCCL demonstrated superior performance to the untrained STCCL on measures including FAD, LSE-D, and CSIM, with FAD reduced from 4.301 to 4.169, LSE-D from 9.443 to 9.216, and CSIM from 0.785 to 0.795.

TABLE III
THE PERFORMANCE OF STCCL WITH AND WITHOUT TRAINING

Settings	Methods	FAD↓	LSE-D↓	CSIM↑
Intra-ID	ICface	6.795	10.083	0.775
	Ours untrained	6.743	9.832	0.781
	Ours trained	**6.634**	**9.411**	**0.806**
	NED	2.108	9.454	0.831
	Ours untrained	1.488	9.421	0.864
	Ours trained	**1.077**	**9.335**	**0.914**
Cross-ID	ICface	9.540	11.238	0.688
	Ours untrained	9.512	10.876	0.691
	Ours trained	**9.445**	**10.311**	**0.730**
	NED	4.448	9.906	0.773
	Ours untrained	4.301	9.443	0.785
	Ours trained	**4.169**	**9.216**	**0.795**

This indicates that by utilizing paired data to train the STCCL, we can more effectively capture the spatial-temporal coherent correlations between the SPFEM's input and output, thereby enhancing the model's guidance. Here, we also present some visual comparison results as shown in Fig. 9. After training with paired data, STCCL can better supervise the generated results of the SPFEM model.

2) Analysis of Spatial Coherent Correlation Learning: The SCC metric and TCC metric respectively supervise the generation of the SPFEM model in the spatial and temporal dimensions. In this section and section V-D3, we analyze their respective roles. We integrate the SCC metric (Ours SCC-only), TCC metric (Ours TCC-only) and STCCL (Ours) into the training of the SPFEM model, with the results shown in the Tables IV and V.

The SCC metric aligns the visual correlation between the input and output of the SPFEM model in the spatial dimension. It ensures that the visual correlations of adjacent local regions within the SPFEM model's input closely resemble those of corresponding regions in the model's output. As shown in Table IV, after removing the TCC metric, Ours SCC-only still shows significant improvements over the Baseline in all three metrics. For example, in the NED baseline and Intra-ID setting, FAD decreased from 2.108 to 1.229, LSE-D decreased from

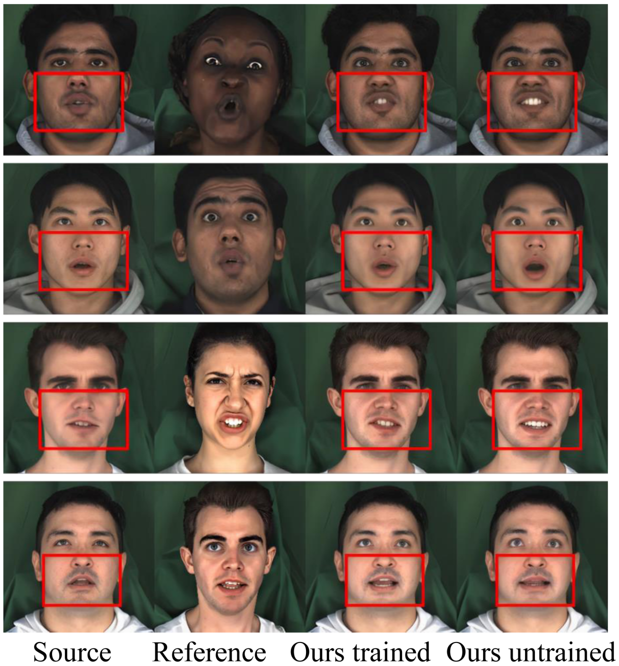

Fig. 9. Several examples are generated by Ours STCCL base on NED baseline, with / without using paired data to train.

TABLE IV
COMPARISON RESULTS OF AVERAGE FAD, CSIM, AND LSE-D FOR ICFACE AND NED WITH THE PROPOSED ALGORITHM, BOTH WITH AND WITHOUT THE TCC METRIC

Settings	Methods	FAD↓	LSE-D↓	CSIM↑
Intra-ID	ICface	6.795	10.083	0.775
	Ours SCC-only	6.691	9.611	0.781
	Ours	**6.634**	**9.411**	**0.806**
	NED	2.108	9.454	0.831
	Ours SCC-only	1.229	9.402	0.885
	Ours	**1.077**	**9.335**	**0.914**
Cross-ID	ICface	9.540	11.238	0.688
	Ours SCC-only	9.490	10.672	0.712
	Ours	**9.445**	**10.311**	**0.730**
	NED	4.448	9.906	0.773
	Ours SCC-only	4.307	9.330	0.786
	Ours	**4.169**	**9.216**	**0.795**

9.454 to 9.402, and CSIM increased from 0.831 to 0.885. We also observed that integrating the SCC metric into the SPFEM model results in noticeable improvements in the FAD metric across different baselines and settings. Empirically, we believe that the SCC metric provides spatial dimension supervision to the SPFEM model, considering the unique two-dimensional organizational structure of images during the supervision process. Consequently, the quality of the generated images is relatively better. Moreover, when the TCC metric and SCC metric are integrated together into the supervision process of the SPFEM model, there are significant improvements in all three metrics compared to the baseline and Ours SCC-only.

3) Analysis of Temporal Coherent Correlation Learning: The TCC metric aligns the visual correlation between the input sequence and the generated sequence of the SPFEM model in the temporal dimension. It ensures that the visual correlations of specific regions across adjacent image frames within the SPFEM model's input sequence are similar to those in the corresponding regions of the model's output sequence. As indicated in Table V, removing the SCC metric, Ours TCC-only still shows significant improvements over the Baseline in all three metrics. Additionally, we observed certain improvements in the LSE-D metric for Ours TCC-only compared to the baseline across different methods and settings. We believe that this is possible because the TCC metric leverages inter-frame prior information in the input sequence. This allows the SPFEM model to consider the motion trajectory of specific regions in the input sequence during the generation process, aligning the mouth shape changes in the input sequence with the generated sequence, thereby improving the LSE-D metric.

TABLE V
COMPARISON RESULTS OF AVERAGE FAD, CSIM, AND LSE-D FOR ICFACE AND NED WITH THE PROPOSED ALGORITHM, BOTH WITH AND WITHOUT THE SCC METRIC

Settings	Methods	FAD↓	LSE-D↓	CSIM↑
Intra-ID	ICface	6.795	10.083	0.775
	Ours TCC-only	6.694	9.512	0.778
	Ours	**6.634**	**9.411**	**0.806**
	NED	2.108	9.454	0.831
	Ours TCC-only	1.342	9.390	0.908
	Ours	**1.077**	**9.335**	**0.914**
Cross-ID	ICface	9.540	11.238	0.688
	Ours TCC-only	9.520	10.513	0.709
	Ours	**9.445**	**10.311**	**0.730**
	NED	4.448	9.906	0.773
	Ours TCC-only	4.318	9.274	0.784
	Ours	**4.169**	**9.216**	**0.795**

TABLE VI
ABLATION STUDY ON THE EFFECT OF THE CORRELATION-AWARE ADAPTIVE STRATEGY (CAAS)

Settings	Methods	FAD↓	LSE-D↓	CSIM↑
Intra-ID	ICface	6.795	10.083	0.775
	Ours w/o CAAS	6.681	9.438	0.799
	Ours CAAS	**6.634**	**9.411**	**0.806**
	NED	2.108	9.454	0.831
	Ours w/o CAAS	1.121	9.369	0.911
	Ours CAAS	**1.077**	**9.335**	**0.914**
Cross-ID	ICface	9.540	11.238	0.688
	Ours w/o CAAS	9.489	10.382	0.724
	Ours CAAS	**9.445**	**10.311**	**0.730**
	NED	4.448	9.906	0.773
	Ours w/o CAAS	4.228	9.227	0.792
	Ours CAAS	**4.169**	**9.216**	**0.795**

4) Analysis of Correlation-Aware Adaptive Strategy: Through empirical analysis, we observed that visual correlations are sensitive to regions. We employed CAAS to direct the STCCL's attention toward those challenging regions, thereby enhancing the consistency of mouth movements between input and output. In this section, we examined the influence of incorporating CAAS into the outcome. As depicted in Table VI, in the absence of CAAS, our approach outperforms NED across all metrics. Moreover, when CAAS is integrated into our method, STCCL achieves even more optimal results by dynamically adapting the learning strategy to different regions.

VI. Limitations and Future Work

The proposed STCCL is a model-agnostic supervisory module that effectively preserves structural dynamics and lip synchronization through spatial-temporal correlation consistency. However, since STCCL prioritizes speech-driven structural alignment over explicit emotion semantics, limitations may arise in challenging scenarios such as extreme emotional intensities or complex cross-identity transfers. Furthermore, the overall performance remains inherently bounded by the generative capacity and domain biases of the underlying backbone models. To address these limitations, future research will focus on extending this framework with explicit emotion-aware constraints. One promising direction involves leveraging limited paired data to learn emotion-specific correlation patterns, which could complement the current speech-driven structural supervision. By jointly modeling fine-grained articulatory consistency and semantic emotion alignment, we aim to further enhance the controllability and realism of high-fidelity emotional talking head generation.

VII. Conclusion

In this work, we uncover that a single speaker articulating identical content across varied emotional states exhibits strong inherent visual correlations in both spatial and temporal dimensions. To capitalize on this insight, we propose the Spatial-Temporal Coherent Correlation Learning (STCCL) algorithm to explicitly model these correlations and integrate them as a universal supervision signal into speech-preserving emotion generation frameworks. By learning spatial and temporal coherent correlation metrics, STCCL aligns local facial regions and ensures consistency across diverse emotional expressions. To address the varying complexity of facial dynamics, we further develop a correlation-aware adaptive strategy that prioritizes challenging regions. Implemented as a plug-and-play loss, STCCL can be seamlessly integrated into diverse generative paradigms, including GAN-based and diffusion-based models. Extensive experiments across foundational SPFEM methods and state-of-the-art talking head frameworks, supported by quantitative analysis, qualitative comparisons, and user studies, demonstrate its effectiveness and robust generalization capability.

References

[1] S. Tripathy, J. Kannala, and E. Rahtu, "ICFace: Interpretable and controllable face reenactment using GANs," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2020, pp. 3385–3394.

[2] M. C. Doukas, M. R. Koujan, V. Sharmanska, A. Roussos, and S. Zafeiriou, "Head2Head : Deep facial attributes re-targeting," *IEEE Trans. Biom., Behav., Ident. Sci.*, vol. 3, no. 1, pp. 31–43, Jan. 2021.

[3] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2017, pp. 2223–2232.

[4] F. P. Papantoniou, P. P. Filntisis, P. Maragos, and A. Roussos, "Neural emotion director: Speech-preserving semantic control of facial expressions in "in-the-wild" videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*2022, pp. 18781–18790.

[5] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4401–4410.

[6] M. Pantic and L. J. M. Rothkrantz, "Automatic analysis of facial expressions: The state of the art," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 12, pp. 1424–1445, Dec. 2000.

[7] Y. Ma et al., "Styletalk: One-shot talking head generation with controllable speaking styles," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 1896–1904.

[8] Y. Gan, Z. Yang, X. Yue, L. Sun, and Y. Yang, "Efficient emotional adaptation for audio-driven talking-head generation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 22634–22645.

[9] W. Tan et al., "Disentangle identity, cooperate emotion: Correlation-aware emotional talking portrait generation," in *Proc. ACM Int. Conf. Multimedia*, 2025, pp. 9987–9995.

[10] T. Chen, J. Lin, Z. Yang, C. Qing, and L. Lin, "Learning adaptive spatial coherent correlations for speech-preserving facial expression manipulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 7267–7276.

[11] Y. Dalva, H. Pehlivan, O. I. Hatipoglu, C. Moran, and A. Dundar, "Image-to-image translation with disentangled latent vectors for face editing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, pp. 14777–14788, Dec. 2023.

[12] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*2017, pp. 1125–1134.

[13] Y. Choi, M. Choi, M. Kim, J.-W. Ha, S. Kim, and J. Choo, "StarGAN: Unified generative adversarial networks for multi-domain image-to-image translation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8789–8797.

[14] Y. Liu, Q. Li, Q. Deng, Z. Sun, and M. Yang, "GAN-based facial attribute manipulation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14590–14610, Dec. 2023.

[15] H. Ding, K. Sricharan, and R. Chellappa, "ExprGAN: Facial expression editing with controllable expression intensity," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 6781–6788.

[16] Z. Geng, C. Cao, and S. Tulyakov, "3D guided fine-grained face manipulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9821–9830.

[17] A. Tewari et al., "StyleRig: Rigging styleGAN for 3D control over portrait images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 6142–6151.

[18] S. d'Apolito, D. P. Paudel, Z. Huang, A. Romero, and L. V. Gool, "GANmut: Learning interpretable conditional space for gamut of emotions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 568–577.

[19] Y. Xu, Z. Yang, T. Chen, K. Li, and C. Qing, "Progressive transformer machine for natural character reenactment," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 19, no. 2s, pp. 1–22, 2023.

[20] A. Pumarola, A. Agudo, A. M. Martinez, A. Sanfeliu, and F. Moreno-Noguer, "Ganimation: One-shot anatomically consistent facial animation," *Int. J. Comput. Vis.*, vol. 128, pp. 698–713, 2020.

[21] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of styleGAN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8110–8119.

[22] R. Abdal, Y. Qin, and P. Wonka, "Image2StyleGAN: How to embed images into the styleGAN latent space," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4432–4441.

[23] R. Abdal, Y. Qin, and P. Wonka, "Image2StyleGAN : How to edit the embedded images?," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8296–8305.

[24] E. Richardson et al., "Encoding in style: A StyleGAN encoder for image-to-image translation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2287–2296.

[25] Y. Alaluf, O. Patashnik, and D. Cohen-Or, "Restyle: A residual-based styleGAN encoder via iterative refinement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 6711–6720.

[26] J. Zhu, Y. Shen, Y. Xu, D. Zhao, Q. Chen, and B. Zhou, "In-domain GAN inversion for faithful reconstruction and editability," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 2607–2621, May 2024.

[27] X. Hu et al., "Style transformer for image inversion and editing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11337–11346.

[28] B. Li et al., "ReGANIE: Rectifying GAN inversion errors for accurate real image editing," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 1269–1277.

[29] X. Yang, X. Xu, and Y. Chen, "Out-of-domain GAN inversion via invertibility decomposition for photo-realistic human face manipulation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 7492–7501.

[30] T. Wang, Y. Zhang, Y. Fan, J. Wang, and Q. Chen, "High-fidelity GAN inversion for image attribute editing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11379–11388.

[31] W. Xia, Y. Zhang, Y. Yang, J.-H. Xue, B. Zhou, and M.-H. Yang, "GAN inversion: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3121–3138, Mar. 2023.

[32] D. Roich, R. Mokady, A. H. Bermano, and D. Cohen-Or, "Pivotal tuning for latent-based editing of real images," *ACM Trans. Graph.*, vol. 42, no. 1, pp. 1–13, 2022.

[33] R. Tzaban, R. Mokady, R. Gal, A. Bermano, and D. Cohen-Or, "Stitch it in time: GAN-based facial editing of real videos," in *Proc. SIGGRAPH Asia 2022 Conf. Papers*, 2022, pp. 1–9.

[34] Y. Xu, B. AlBahar, and J.-B. Huang, "Temporally consistent semantic video editing," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 357–374.

[35] Y. Xu, S. He, K.-Y. K. Wong, and P. Luo, "Rigid: Recurrent GAN inversion and editing of real face videos," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 13691–13701.

[36] X. Ji et al., "EAMM: One-shot emotional talking face via audio-based emotion-aware motion model," in *Proc. SIGGRAPH Conf.*, 2022, pp. 1–10.

[37] B. Liang et al., "Expressive talking head generation with granular audio-visual control," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 3387–3396.

[38] S. Tan, B. Ji, M. Bi, and Y. Pan, "EDTalk: Efficient disentanglement for emotional talking head synthesis," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 398–416.

[39] K. Wang et al., "MEAD: A large-scale audio-visual dataset for emotional talking-face generation," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K., 2020, pp. 700–717.

[40] S. E. Eskimez, Y. Zhang, and Z. Duan, "Speech driven talking face generation from a single image and an emotion condition," *IEEE Trans. Multimedia*, vol. 24, pp. 3480–3490, 2022.

[41] L. Tian, Q. Wang, B. Zhang, and L. Bo, "EMO: Emote portrait alive generating expressive portrait videos with audio2video diffusion model under weak conditions," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 244–260.

[42] S. Shen et al., "Difftalk: Crafting diffusion models for generalized audio-driven portraits animation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*2023, pp. 1982–1991.

[43] I. Magnusson, A. Sankaranarayanan, and A. Lippman, "Invertible Frowns: Video-to-video facial emotion translation," in *Proc. ADGD Workshop ACM Multimedia 2021*, 2021, pp. 25–33.

[44] V. Blanz and T. Vetter, "A morphable model for the synthesis of 3D faces," in *Proc. 26th Annu. Conf. Comput. Graph. Interactive Techn.*, 2023, pp. 157–164.

[45] Y. Feng, H. Feng, M. J. Black, and T. Bolkart, "Learning an animatable detailed 3D face model from in-the-wild images," *ACM Trans. Graph.*, vol. 40, no. 4, pp. 1–13, 2021.

[46] Z. Ding, X. Zhang, Z. Xia, L. Jebe, Z. Tu, and X. Zhang, "Diffusion-Rig: Learning personalized priors for facial appearance editing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*2023, pp. 12736–12746.

[47] H. Li, H. Chen, Y. Huang, T. Chen, and S. Huang, "Enhancing lip dynamic authenticity: Learning 3D temporal representations for talking head synthesis," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 21, pp. 1–21, 2025.

[48] Z. Sun et al., "Continuously controllable facial expression editing in talking face videos," *IEEE Trans. Affev. Comput.*, vol. 15, no. 1, pp. 1400–1413, Third Quarter 2024.

[49] G. K. Solanki and A. Roussos, "Deep semantic manipulation of facial videos," in *Proc. Eur. Conf. Comput. Vis.*, 2023, pp. 104–120.

[50] Z. Wu, Z. Zhu, J. Du, and X. Bai, "CCPL: Contrastive coherence preserving loss for versatile style transfer," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 189–206.

[51] Y. Deng, F. Tang, W. Dong, H. Huang, C. Ma, and C. Xu, "Arbitrary video style transfer via multi-channel correlation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 1210–1217.

[52] T. Chen et al., "Dynamic correlation learning and regularization for multi-label confidence calibration," *IEEE Trans. Image Process.*, vol. 33, pp. 4811–4823, 2024.

[53] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ARCFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4690–4699.

[54] V. Blanz and T. Vetter, "A morphable model for the synthesis of 3D faces," in *Proc. Annu. Conf. Comput. Graph. Interactive Techn.*, 1999, pp. 187–194.

[55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*2016, pp. 770–778.

[56] T. Chen, J. Lin, Z. Yang, C. Qing, Y. Shi, and L. Lin, "Contrastive decoupled representation learning and regularization for speech-preserving facial expression manipulation," *Int. J. Comput. Vis.*, vol. 133, pp. 3822–3838, 2025.

[57] J. Lin et al., "Neural scene designer: Self-styled semantic image manipulation," *IEEE Trans. Image Process.*, vol. 34, pp. 6577–6588, 2025.

[58] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[59] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.

[60] S. R. Livingstone and F. A. Russo, "The Ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLoS One*, vol. 13, no. 5, 2018, Art. no. e0196391.

[61] K. R. Prajwal, R. Mukhopadhyay, V. P. Namboodiri, and C. V. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild," in *Proc. ACM Int. Conf. Multimedia*, 2020, pp. 484–492.

[62] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild," in *Proc. Asian Conf. Comput. Vis.*, 2016, pp. 251–263.

Tianshui Chen received the BE degree from the School of Information and Science Technology, in 2013, the PhD degree in computer science from the School of Data and Computer Science Sun Yat-sen University, Guangzhou, China, in 2018. He is currently an associate professor with the Guangdong University of Technology. He has authored and coauthored more than 60 papers published in top-tier academic journals and conferences, including *IEEE Transactions on Pattern Analysis and Machine Intelligence*, IJCV, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Multimedia*, CVPR, ICCV, AAAI, IJCAI, and ACM MM. His research interests include artificial intelligence, multimodal large models, and generative AI. He was a reviewer for numerous academic journals and conferences. He was the recipient of the Best Paper Diamond Award at IEEE ICME 2017.

Jianman Lin received the BSc degree from the Guangdong University of Technology, Guangzhou, China, in 2024. He is currently working toward the master's degree in electronic information with the South China University of Technology. He has authored or coauthored revised papers in prestigious conferences and journals, including CVPR and IJCV. His research interests include artificial intelligence, multimodal large models, and generative AI.

Zhijing Yang received the BS and PhD degrees in mathematics and computing science from Sun Yat-sen University, Guangzhou China, in 2003 and 2008, respectively. Between Jul. 2009 to Dec. 2009, he was a visiting research scholar with the School of Computing, Informatics and Media, University of Bradford, U.K., and a research fellow with the School of Engineering, University of Lincoln, U.K, between Jan. 2011 to Jan. 2013. He is currently a professor and vice dean with the School of Information Engineering, Guangdong University of Technology, China. He has authored or coauthored more than 80 peer-reviewed journal and conference papers, including *IEEE Transactions on Circuits and Systems for Video Technology*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Geoscience and Remote Sensing*, and PR. His research interests include machine learning and pattern recognition.

Chunmei Qing (Member, IEEE) received the BSc degree in information and computation science from Sun Yat-sen University, China, in 2003, and the PhD degree in electronic imaging and media communications from the University of Bradford, U.K., in 2009. She was a postdoctoral researcher with the University of Lincoln, U.K. She is currently a professor with the School of Electronic and Information Engineering, South China University of Technology, Guangzhou, China. Her research interests mainly include image/video processing, computer vision, and affective computing.

Guangrun Wang received the two BE degrees and one PhD degree from Sun Yat-sen University (SYSU), in 2014 and 2020, respectively. He was a research fellow with the Department of Engineering Science, University of Oxford. He is currently a young researcher and associate professor of computer science and AI with SYSU. He is also a National Excellent Young scientist and a Huawei Talent Faculty Fund Laureate, working on new AI architectures. He was a visiting scholar with the Chinese University of Hong Kong (CUHK). He was an area chair for ICLR and IJCAI, as a distinguished senior program committee member for IJCAI, and an outstanding reviewer on six separate conferences for ICLR, NeurIPS, and ICCV. He was the recipient of the Pattern Recognition Best Paper Award in 2018, Top Chinese Rising Stars in Artificial Intelligence Award, and Wu Wen-Jun Best Doctoral Dissertation Award.

Liang Lin (Fellow, IEEE) is currently a full professor with Sun Yat-sen University. From 2008 to 2010, he was a postdoctoral fellow with the University of California, Los Angeles. From 2016 to 2018, he led the SenseTime R&D Teams, to develop cutting-edge and deliverable solutions for computer vision, data analysis and mining, and intelligent robotic systems. He has authored and coauthored more than 100 papers in top-tier academic journals and conferences (e.g., 15 papers in *IEEE Transactions on Pattern Analysis and Machine Intelligence* and IJCV and more than 60 papers in CVPR, ICCV, NIPS, and IJCAI). He was an associate editor for *IEEE Transactions on Human-Machine Systems*, *The Visual Computer*, and *Neurocomputing* and as an area/session chair for numerous conferences, such as CVPR, ICME, ACCV, and ICMR. He was the recipient of the Annual Best Paper Award by Pattern Recognition (Elsevier) in 2018, Best Paper Diamond Award at IEEE ICME 2017, Best Paper Runner-Up Award at ACM NPAR 2010, Google Faculty Award in 2012, Best Student Paper Award at IEEE ICME 2014, and Hong Kong Scholars Award in 2014. He is a fellow of IAPR and IET.